

Zhike three-phase inverter swimming pool heat pump - mainboard communication protocol (2022-03-29)

English translation of document RCDF211268. R = read-only parameter, RW = read/write parameter.

1. Communication rules

1. Communication uses the RS485 bus: asynchronous serial signal with 1 start bit, 8 data bits, 1 stop bit, no parity check, baud rate 9600 bps.
2. Compliant with the standard MODBUS RTU protocol, 16-bit data structure, 16-bit CRC check with the low byte first and the high byte last.
3. Unit slave address #1 to #16, set by DIP switches 1-4.
4. The upper computer is the calling master; the controller is the slave.
5. Master/slave communication uses three commands:

Command 03H (read one or more registers)

Send: [device address] + [command 03H] + [start register address high 8 bits] + [low 8 bits] + [number of registers to read high 8 bits] + [low 8 bits] + [CRC low 8 bits] + [CRC high 8 bits]

Response: [device address] + [command 03H] + [number of bytes returned] + [data 1] + [data 2] + ... + [data n] + [CRC low 8 bits] + [CRC high 8 bits]

Command 06H (modify a single register)

Send: [device address] + [command 06H] + [register address high 8 bits] + [low 8 bits] + [data high 8 bits] + [low 8 bits] + [CRC low 8 bits] + [CRC high 8 bits]

Response: the command sent by the computer is returned unchanged if successful; otherwise no response

Command 10H (modify multiple registers)

Send: [device address] + [command 10H] + [start register address high 8 bits] + [low 8 bits] + [number of registers high 8 bits] + [low 8 bits] + [number of bytes] + [data 1 high 8 bits] + [low 8 bits] + ... + [data N high 8 bits] + [low 8 bits] + [CRC low 8 bits] + [CRC high 8 bits]

Response: [device address] + [command 10H] + [start register address high 8 bits] + [low 8 bits] + [number of registers high 8 bits] + [low 8 bits] + [CRC low 8 bits] + [CRC high 8 bits]

2. Parameter address table

R = read-only, RW = readable and writable.

Address	Description	Setting range	Default / Remarks
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Address	Description	Setting range	Default / Remarks
R 0x0000	Reserved		
R 0x0001	Reserved		
R 0x0002	Control switch status		See flag description
R 0x0003	Operating status flags		See flag description
R 0x0004	Output status flags 1		See flag description
R 0x0005	Output status flags 2		See flag description
R 0x0006	Output status flags 3		See flag description
R 0x0007	Fault flags 1		See flag description
R 0x0008	Fault flags 2		See flag description
R 0x0009	Fault flags 3		See flag description
R 0x000A	Fault flags 4		See flag description
R 0x000B	Fault flags 5		See flag description
R 0x000C	Fault flags 6		See flag description
R 0x000D	Fault flags 7		See flag description
R 0x000E	Reserved		
R 0x000F	Inlet water temperature	-30~99℃	
R 0x0010	Outlet water temperature	-30~99℃	
R 0x0011	Ambient temperature	-30~99℃	
R 0x0012	Coil 1 temperature	-30~99℃	
R 0x0013	Suction gas temperature 1	-30~99℃	
R 0x0014	Cooling coil 1 temperature	-30~99℃	
R 0x0015	Discharge temperature 1	0~125℃	
R 0x0016	Reserved		

Address	Description	Setting range	Default / Remarks
R 0x0017	Reserved		
R 0x0018	Main EXV opening 1		
R 0x0019	Cooling EXV opening 1		
R 0x001A	Compressor 1 target frequency		
R 0x001B	Compressor 1 actual frequency		
R 0x001C	Inverter module 1 fault code 1		
R 0x001D	Inverter module 1 fault code 2		
R 0x001E	DC bus voltage 1		
R 0x001F	Reserved		
R 0x0020	Reserved		
R 0x0021	Reserved		
R 0x0022	Reserved		
R 0x0023	Reserved		
R 0x0024	DC fan 1 speed		
R 0x0025	DC fan 2 speed		
R 0x0026	Coil 2 temperature	-30~99℃	
R 0x0027	Suction gas temperature 2	-30~99℃	
R 0x0028	Cooling coil 2 temperature	-30~99℃	
R 0x0029	Discharge temperature 2	0~125℃	
R 0x002A	Reserved		
R 0x002B	Reserved		
R 0x002C	Main EXV opening 2		

Address	Description	Setting range	Default / Remarks
R 0x002D	Cooling EXV opening 2		
R 0x002E	Compressor 2 target frequency		
R 0x002F	Compressor 2 actual frequency		
R 0x0030	Inverter module 2 fault code 1		
R 0x0031	Inverter module 2 fault code 2		
R 0x0032	DC bus voltage 2		
R 0x0033	Reserved		
R 0x0034	Reserved		
R 0x0035	Reserved		
R 0x0036	Reserved		
R 0x0037	Reserved		
R 0x0038	Reserved		
R 0x0039	Reserved		
R 0x003A	Reserved		
R 0x003B	Reserved		
R 0x003C	Reserved		
R 0x003D	Reserved		
R 0x003E	Reserved		
RW 0x003F	Parameter flag definition		See flag description
RW 0x0040	Mode		See flag description
RW 0x0041	Heating temperature setpoint	8℃~40℃	27℃

Address	Description	Setting range	Default / Remarks
RW 0x0042	Cooling temperature setpoint	8℃~28℃	27℃
RW 0x0043	Manual frequency setting		
RW 0x0044	Manual EXV step position	20~450	300
RW 0x0045	Manual auxiliary-valve step position		
RW 0x0046	Manual frequency setting 2		
RW 0x0047	Manual EXV 2 step position		
RW 0x0048	Manual auxiliary-valve 2 step position		
RW 0x0049	Manual fan speed		
RW 0x004A	Auto-mode temperature setpoint		
RW 0x004B	Reserved		
RW 0x004C	Mode changeover time	3~30min	10 min
RW 0x004D	Temperature hysteresis (differential)	1℃~18℃	1℃
RW 0x004E	Inlet water temperature compensation		
RW 0x004F	Reserved		
RW 0x0050	Defrost interval	20MIN~90MIN	40MIN
RW 0x0051	Defrost start temperature	-15℃~-1℃	-6℃

Address	Description	Setting range	Default / Remarks
RW 0x0052	Defrost duration	5MIN~20MIN	11MIN
RW 0x0053	Defrost termination temperature	1℃~40℃	16℃
RW 0x0054	Defrost ambient-to-coil temperature difference	0℃~15℃	6℃
RW 0x0055	Defrost ambient temperature threshold	0℃~20℃	17℃
RW 0x0056	EXV adjustment period	20S~90S	30S
RW 0x0057	Heating target superheat	-5℃~10℃	1℃
RW 0x0058	EXV discharge-temperature target	70℃~125℃	88℃
RW 0x0059	EXV opening during defrost	20~450	400
RW 0x005A	Minimum EXV opening	50~150	80
RW 0x005B	Cooling target superheat	-5℃~10℃	1℃
RW 0x005C	Reserved		
RW 0x005D	Frequency setting F1	30~90Hz	40Hz
RW 0x005E	Frequency setting F2	30~90Hz	44Hz
RW 0x005F	Frequency setting F3	30~90Hz	48Hz
RW 0x0060	Frequency setting F4	30~90Hz	54Hz
RW 0x0061	Frequency setting F5	30~90Hz	58Hz

Address	Description	Setting range	Default / Remarks
RW 0x0062	Frequency setting F6	30~90Hz	64Hz
RW 0x0063	Frequency setting F7	30~90Hz	72Hz
RW 0x0064	Frequency setting F8	30~90Hz	80Hz
RW 0x0065	Frequency setting F9	30~90Hz	84Hz
RW 0x0066	Frequency setting F10	30~90Hz	90Hz
RW 0x0067	Discharge temperature setting TP0	50~125℃	95℃
RW 0x0068	Discharge temperature setting TP1	50~125℃	100℃
RW 0x0069	Discharge temperature setting TP2	50~125℃	105℃
RW 0x006A	Discharge temperature setting TP3	50~125℃	110℃
RW 0x006B	Discharge temperature setting TP4	50~125℃	115℃
RW 0x006C	Reserved		
RW 0x006D	Reserved		
RW 0x006E	Reserved		
RW 0x006F	Fan speed level 1	20~100	46
RW 0x0070	Fan speed level 2	20~100	52
RW 0x0071	Fan speed level 3	20~100	58

Address	Description	Setting range	Default / Remarks
RW 0x0072	Fan speed level 4	20~100	64
RW 0x0073	Fan speed level 5	20~100	72
RW 0x0074	Fan speed level 6	20~100	85
RW 0x0075	Fan speed level selection	0~6	0
RW 0x0076	Fan type (0 = AC / 1 = DC / 2 = EC)	0=AC/1=DC/2=EC Fan Modle	1
RW 0x0077	Reserved		
RW 0x0078	Reserved		
RW 0x0079	Timer enable flag		
RW 0x007A	Timer 1 ON hour	0-23H	
RW 0x007B	Timer 1 ON minute	0-59MIN	
RW 0x007C	Timer 1 OFF hour	0-23H	
RW 0x007D	Timer 1 OFF minute	0-59MIN	
RW 0x007E	Timer 2 ON hour	0-23H	
RW 0x007F	Timer 2 ON minute	0-59MIN	
RW 0x0080	Timer 2 OFF hour	0-23H	
RW 0x0081	Timer 2 OFF minute	0-59MIN	

Address	Description	Setting range	Default / Remarks
RW 0x0082			See flag description

3. Flag descriptions

Control switch port (0x0082)

Bit	Description
bit 0	Reserved
bit 1	Reserved
bit 2	High pressure switch
bit 3	Reserved
bit 4	Low pressure switch
bit 5	Low pressure switch 2
bit 6	Water flow switch
bit 7	High pressure switch 2

Work status flag (0x0003)

Bit	Description
bit 0	Hot water available
bit 1	High temperature
bit 2	Heating
bit 3	Cooling
bit 4	Reserved
bit 5	High water level
bit 6	Low water level
bit 7	Defrost

Output flag 1 (0x0004)

Bit	Description
bit 0	Compressor 1
bit 1	Compressor 2
bit 2	Reserved
bit 3	Reserved
bit 4	Reserved
bit 5	Fan
bit 6	Four-way valve
bit 7	High / low fan speed

Output flag 2 (0x0005)

Bit	Description
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bit 0	Reserved
bit 1	Reserved
bit 2	Reserved
bit 3	Make-up water valve
bit 4	Reserved
bit 5	Reserved
bit 6	Air-conditioning cooling demand
bit 7	Air-conditioning heating demand

Output flag 3 (0x0006)

Bit	Description
bit 0	Reserved
bit 1	Crankcase electric heater
bit 2	Circulating water pump
bit 3	Reserved
bit 4	Reserved
bit 5	Reserved
bit 6	Reserved
bit 7	Reserved

Fault flag 1 (0x0007)

Bit	Description
bit 0	Reserved
bit 1	Ambient temperature fault
bit 2	Coil 1 temperature fault
bit 3	Reserved
bit 4	Outlet water temperature fault
bit 5	High pressure fault
bit 6	Low pressure fault
bit 7	Reserved

Fault flag 2 (0x0008)

Bit	Description
bit 0	Water flow switch fault
bit 1	Reserved
bit 2	Heating outlet water over-temperature protection
bit 3	Reserved
bit 4	Reserved
bit 5	Coil 2 temperature fault
bit 6	High pressure switch 2 fault
bit 7	Low pressure switch 2 fault

Fault flag 3 (0x0009)

Bit	Description
bit 0	Reserved
bit 1	Reserved
bit 2	Reserved
bit 3	Reserved
bit 4	Reserved
bit 5	DC fan 2 fault
bit 6	Exhaust 1 temperature fault
bit 7	Reserved

Fault flag 4 (0x000A)

Bit	Description
bit 0	Inlet water temperature fault
bit 1	Exhaust 1 over-temperature fault
bit 2	Reserved
bit 3	Reserved
bit 4	Reserved
bit 5	Cooling outlet water over-cooling protection
bit 6	Return gas 1 temperature fault
bit 7	Reserved

Fault flag 5 (0x000B)

Bit	Description
bit 0	Reserved
bit 1	Reserved
bit 2	Coil 1 over-temperature protection
bit 3	Cooling coil 1 temperature fault
bit 4	Reserved
bit 5	Coil 2 over-temperature protection
bit 6	Exhaust 2 over-temperature fault
bit 7	Reserved

Fault flag 6 (0x000C)

Bit	Description
bit 0	Reserved
bit 1	Reserved
bit 2	Reserved
bit 3	Reserved
bit 4	Second-stage anti-freeze
bit 5	First-stage anti-freeze

bit 6	Reserved
bit 7	Reserved

Fault flag 7 (0x000D)

Bit	Description
bit 0	Cooling coil 2 temperature fault
bit 1	Reserved
bit 2	Inverter module 2 communication fault
bit 3	Return gas 2 temperature fault
bit 4	Inverter module 1 communication fault
bit 5	Exhaust 2 temperature fault
bit 6	DC fan 1 fault
bit 7	Reserved

Parameter flag definition 1 (0x003F)

Bit	Description
bit 0	Wire controller ON/OFF
bit 1	Manual frequency
bit 2	Expansion valve mode selection: 0 = manual / 1 = automatic
bit 3	Reserved
bit 4	Water pump working mode: 0 = continuous / 1 = periodic operation
bit 5	Cooling expansion valve working mode: 0 = ambient / 1 = superheat
bit 6	Silent mode: 0 = off / 1 = on
bit 7	Reserved
bit 12	Reserved
bit 13	Fan mode
bit 14	Forced defrost
bit 15	Reserved

Mode (0x0040)

Bit	Description
bit 0	Mode selection: 1 = heating / 2 = cooling / 7 = auto
bit 1	Reserved
bit 2	Reserved
bit 3	Reserved
bit 4	Boost (powerful) mode: 0 = off / 1 = on
bit 5	Reserved
bit 6	Reserved
bit 7	Reserved

